CCS2305:WEB PROGRAMMING

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COURSE SPECS

Course Code (Title)	CCS2305 (Web Programming)
Level/Semester	2/Spring (2023-2024)
#CH **#Hours (Lecture)** **#Hours (Lab)**	3 2 2
Lecturer	Dr. Nashwa Abdelghaffar
Contact	

LOGISTICS

Course Code (Title)	CCS2305 (Web Programming)
Grading: 7^{th} Week Written Quiz LAB Quiz	 30 % 15 % 15 %
12^{th} Week	20 % Project
Class Work	10 %
Final Written Exam	40 %

COURSE CONTENT

- Introduction to web programming
- HTML (HyperText Mark-up Language)
- CSS (Cascading Style Sheet)
- JavaScript
- PHP

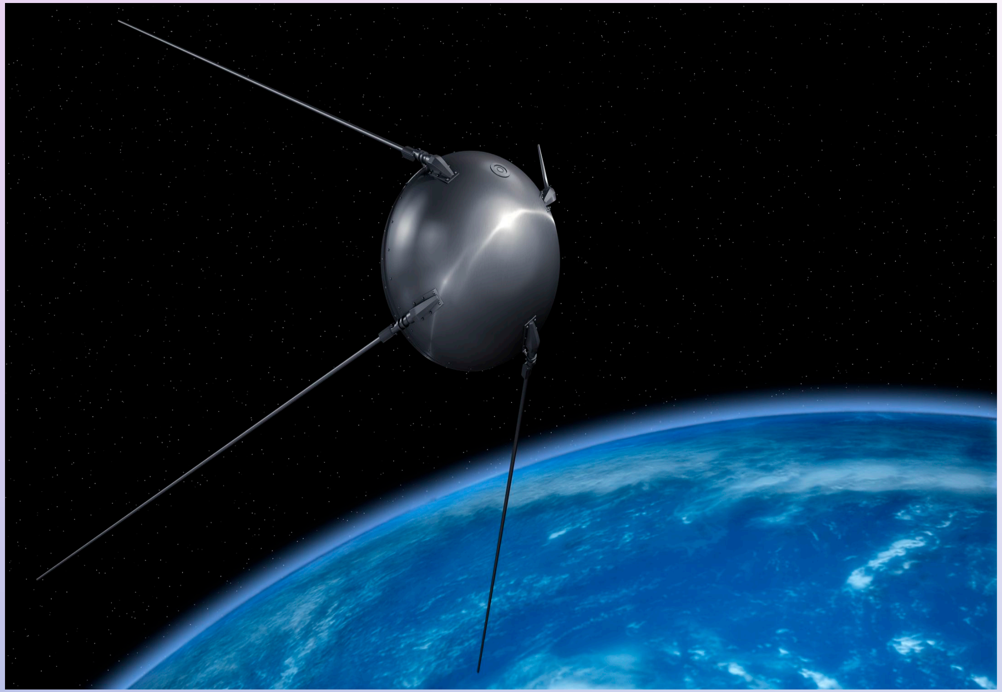
OUTLINE

- History of the Internet
- The World Wide Web
- The Internet vs. the Web
- Common Terms
- Introduction to HTML

HISTORY OF THE INTERNET

- The heating up of the cold war was one of the catalyst in the formation of the Internet.
- The US realized it needed a communications system that could not be affected by a soviet nuclear attack for government researchers to share information.
- This eventually led to the formation of the **ARPANET** (**A**dvanced **R**esearch **P**rojects **A**gency **NET**work).

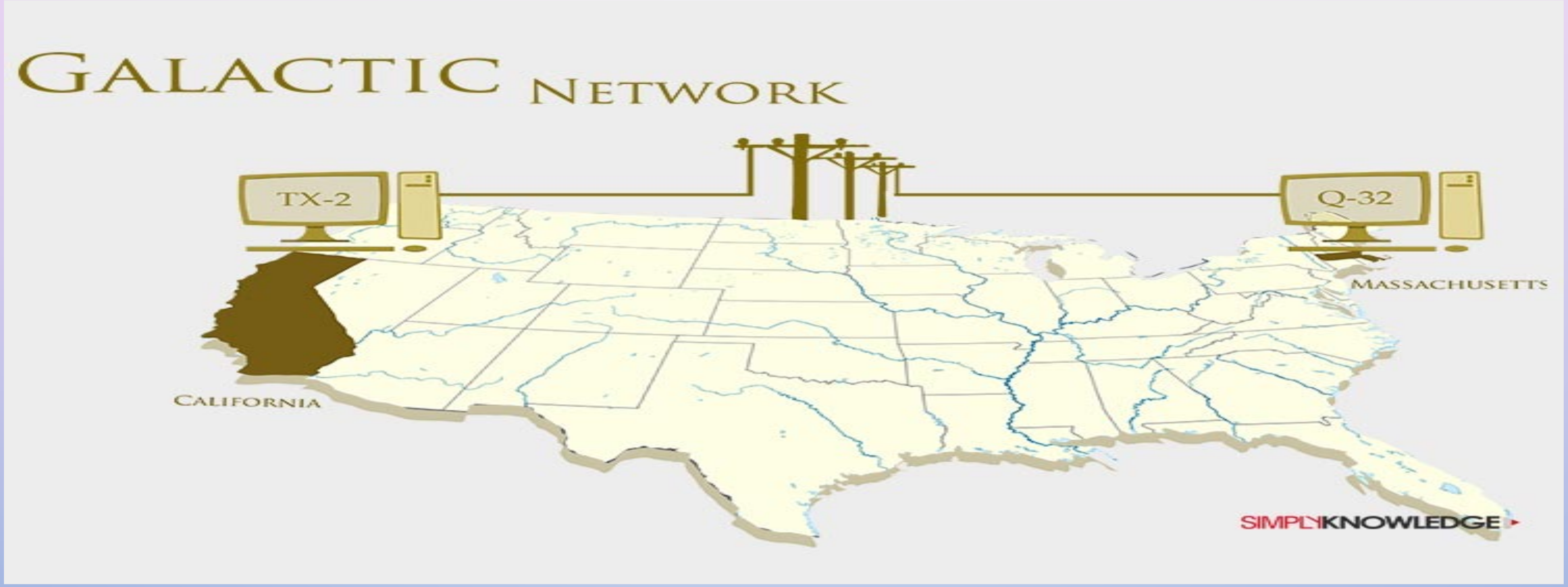
HISTORY OF THE INTERNET



Sputnik Satellite

ARPANET interface

GALACTIC NETWORK



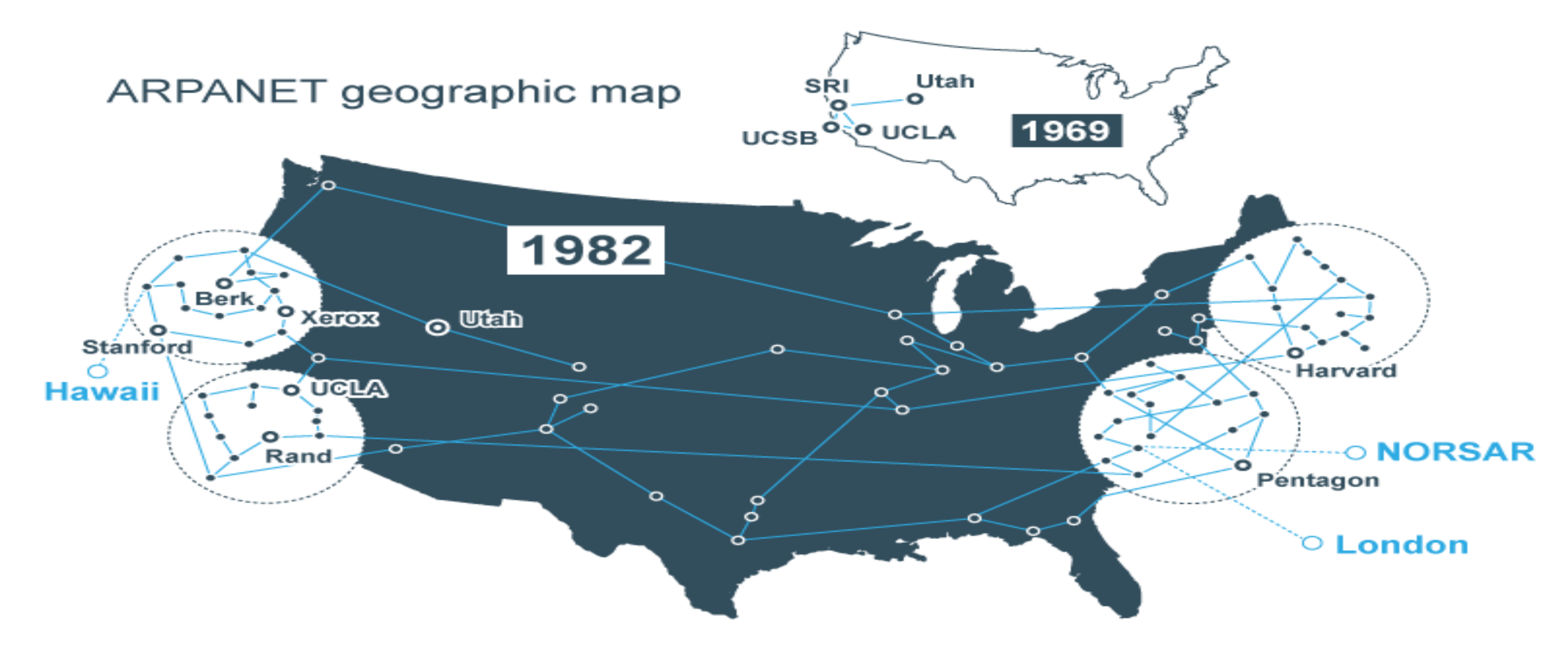
ARPANET

- Its initial purpose was to link computers at pentagon-funded research institutions over telephone lines.
- In the early of 1960s, Baran proposed a communication network with no central command point. If one point was destroyed, all surviving points would still be able to communicate with each other.
- In 1962, Licklider was a major proponent of the importance of the networking concept (Galactic network).
- In 1965, Lawrence Roberts made two separate computers in different places “talk” to each other for the first time.

AN INITIAL ARPANET

- The first message sent over ARPANET happened on Oct. 29, 1969.
- Charley Kline, who was a student at the University of California Los Angeles (**UCLA**), tried to log in to the mainframe at the Stanford Research Institute (**SRI**).
- Two more universities joined ARPANET as founding members on Dec. 5, 1969. These were the **University of California, Santa Barbara** and **University of Utah** school of computing.

GROWTH OF ARPANET



WORLD WIDE WEB

- TIM BERNERS-LEE, a British scientist, invented the world wide web (WWW) in 1989, while working at CERN.
- The basic idea of the www was to merge the evolving technologies of computers, data networks and hypertext into a powerful and easy to use global information system.
- This outlined the principal concepts, and it defined important terms behind the web. The document described a "hypertext project" called ”World Wide Web" in which a "web" of "hypertext documents" could be viewed by “browsers”.

WORLD WIDE WEB

- The fundamental technologies for today's web are as follows.
- **HTML** (HyperText Markup Language): the markup formatting language for the web.
- **URI** (Uniform Resource Identifier): a kind of address that is unique and used to identify each resource on the web.
- **HTTP** (Hyper Text Transfer Protocol): a protocol that allows the retrieval of linked resources across the web.

THE INTERNET VS. THE WEB

- The **Internet** is the largest network in the world that is concerned about communicating computers to each others to share information.
- The **Internet** is the physical network that is used for communicating computers with each other for exchanging information between them.
- The Web is a system of interconnected public webpages accessible through the Internet.
- The Web or **World Wide Web** (www) provides a multimedia interface to resources available on the internet.

WEB BROWSERS

- **Web browsers** are programs that provide access to web resources on the internet.
- These programs connect you to remote computers; open and transfer files; display text, images, and multimedia; and provide in one tool an uncomplicated interface to the internet and web documents.
- Well-known browsers are Apple Safari, Google Chrome, Microsoft Edge, and Mozilla Firefox.

WEB SERVER

- A web server is a computer system capable of delivering web content to end users over the Internet via a web browser.
- The end user processes a request via a web browser installed on a web server. The communication between a web server or browser and the end user takes place using Hyper Text Transfer Protocol (**HTTP**).
- The primary role of a web server is to store, process, and deliver requested information or webpages to end users.

IP ADDRESS

- Every device connected to the internet is given a unique **IP** (Internet Protocol) number, known as an IP address.
- All IP addresses have this format: four sets of numbers separated by three periods. This is called dotted-decimal notation (**192.168.1.2**)
- For example, the IP address is akin to a phone number assigned to a smartphone. **TCP** (Transmission Control Protocol) is the computer networking version of the technology used to make the smartphone ring and enable its user to talk to the person who called them.
- The two protocols (**TCP/IP**) are frequently used together and rely on each other for data to have a destination and safely reach it

DOMAIN NAME SYSTEM (DNS)

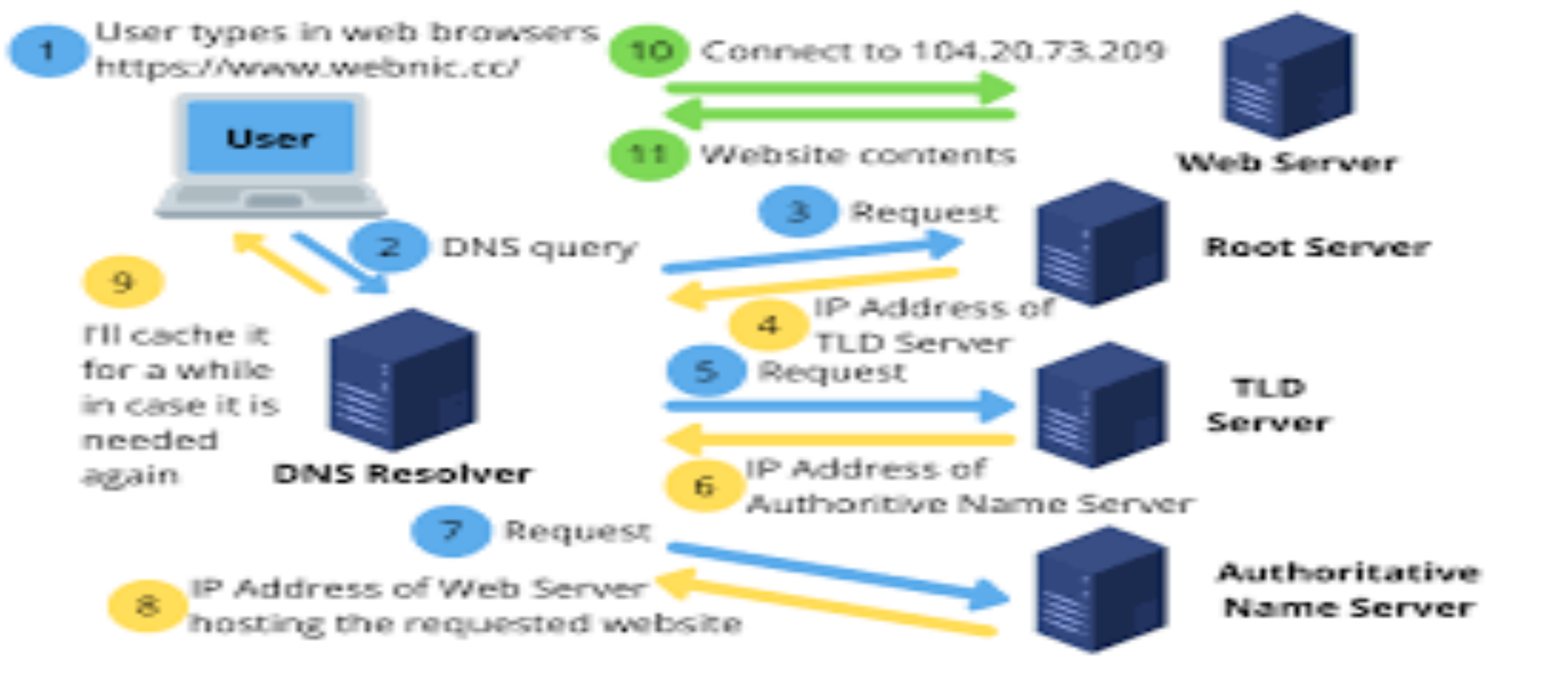
- It is the internet's equivalent of a phone book and converts hard-to-remember IP addresses into simple names.
- There are 4 DNS servers involved in loading a web page:
  - DNS Recursor: is a server designed to receive queries from client machines through applications such as web browsers.
  - Root Server: is the first step in translating human readable host names into IP addresses.
  - The Top-Level Domain Server (TLD): is the next step in the search for a specific IP address, and it hosts the last portion of a hostname.
  - Authoritative Nameserver: is the last stop in the nameserver query that actually holds, and is responsible for, DNS resource records.

WEB ADDRESS

- To access web resources, the address or location of these resources has to specified.
- These addresses are called **Uniform Resource Locators** (URLs) and they have at least two parts.
- The first part identifies the protocol used to connect to the resource for exchanging data between computers
- The second part (domain name) identifies the specific address where the resource is located.

https://www.mathworks.com

DNS & WEB SERVER



WEB PAGES

- Accessing **web resources** through web browsers results in websites or **web pages**, which are document files.
- **Web pages** are stored in **web servers**, which execute a software for providing access to the web pages.
- These documents typically contain **HyperText Markup Language** (HTML), a markup language for displaying web pages.
- The browser interprets the html formatting instructions and displays the document as a web page.
- Interactive and animated websites are provided using one the following technologies: JavaScript, PHP, and CSS.

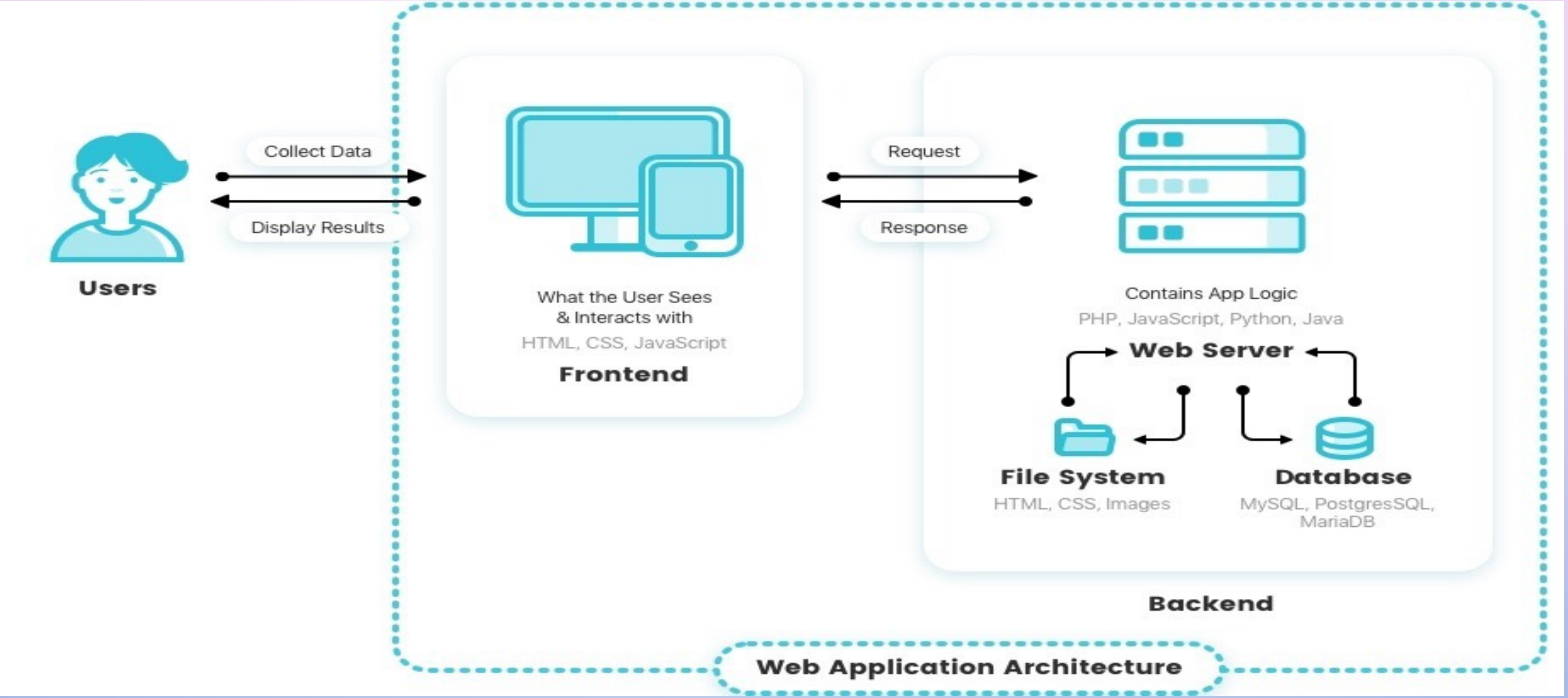
WEB PAGES (CONT.)

- **Static websites** consist of HTML files with fixed content that display the same information to every visitor. In other words, each page you see in the browser is a view of a single HTML file on the server.
- By contrast, **dynamic websites** are generated with backend programming such as PHP. Each page is generated by the application on the fly.
- Dynamic sites access content and data from a database, and the final pages may be customized for each user.

WEB APPLICATION VS. WEBSITE

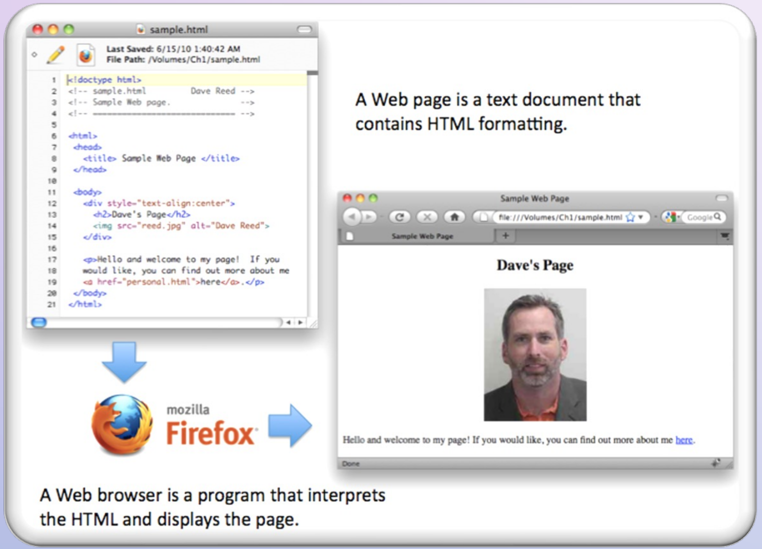
- A website is a collection of digital pages that are accessed through a web browser. A web app is an application that performs specific tasks and is accessed through a web browser.
- Web apps are usually written in a scripting language such as PHP, Perl or Java and use a database such as MYSQL or PostgreSQL to store data.
- The main difference is that a website is static, and a web application is dynamic.

WEB APPLICATION ARCHITECTURE



https://reinventy.com/blog/fundamentals-web-application-architecture/

HTML & WEB PAGES



- Recall that web pages are file documents that typically contain hypertext markup language (HTML), a markup language for displaying web pages.
- The browser interprets the html formatting instructions and displays the documents as web pages.
- Html specifies formatting within a page using tags, where a tag is a word or symbol surrounded by angle brackets (<>).

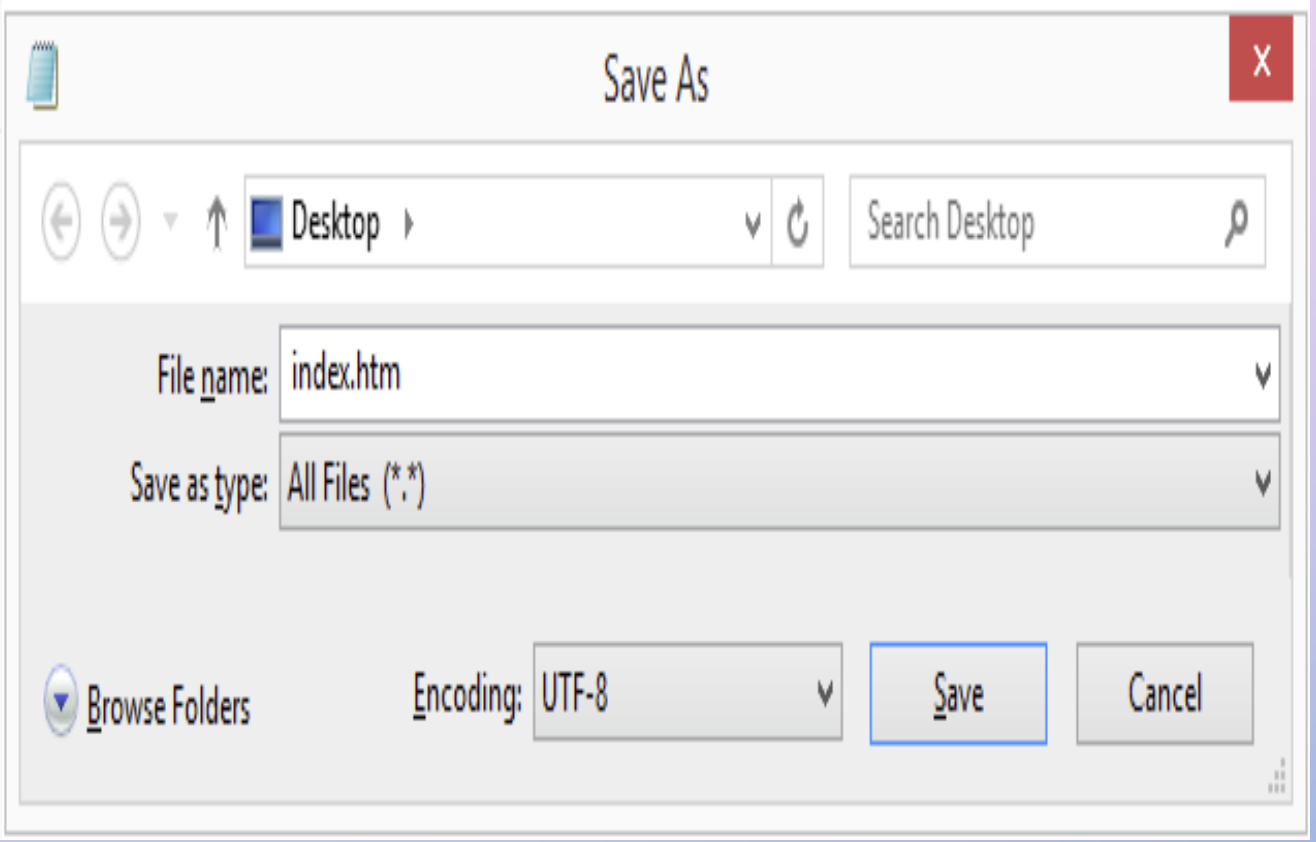
HTML EDITORS

- HTML pages can be created using ANY text editors like notepad, but specialized text editors can be more convenient and have added functionality.

HTML EDITORS (CONT.)

Untitled-11

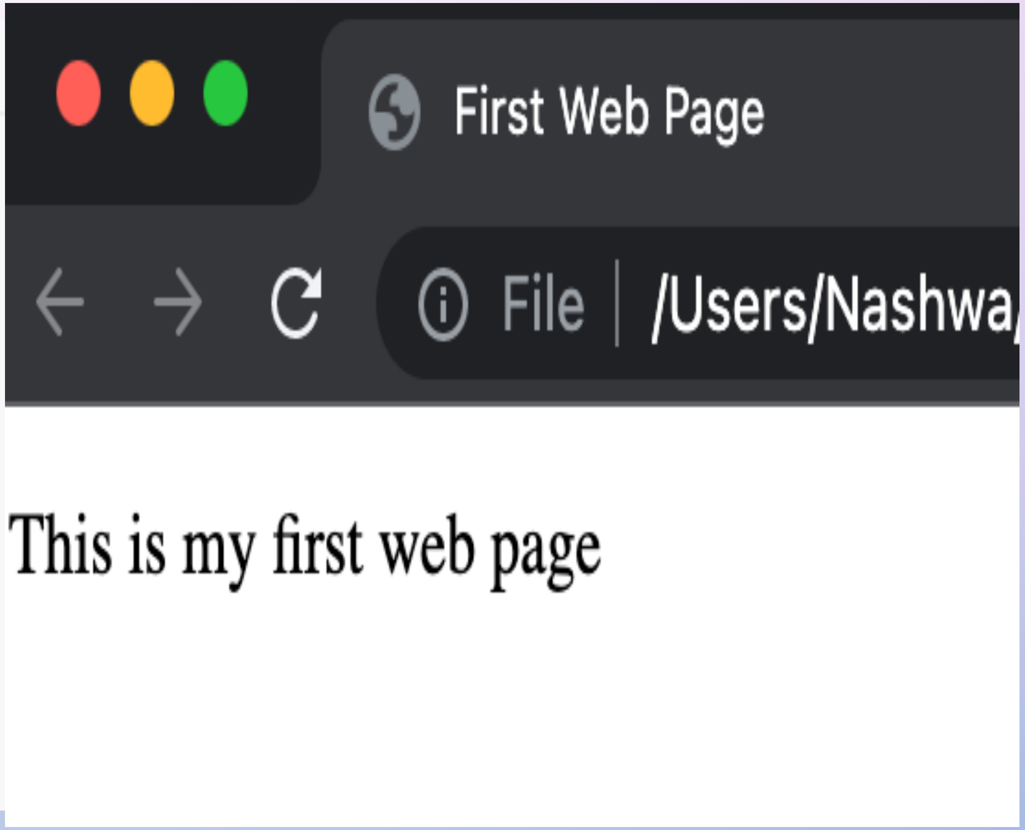
```
<!DOCTYPE html>
<html>
    <head>
        <title> Title here </title>
    </head>
    <body>
        content goes here
    </body>
</html>
```



HTML EDITORS (CONT.)

firstwebpage.html

```html
<!DOCTYPE html>
<html>
    <head>
        <title> First Web Page </title>
    </head>
    <body>
        <p> This is my first web page </p>
    </body>
</html>
```



HTML EDITORS (CONT.)

- Here is a list of some specialized HTML editors evolving from simple text editors:

- Notepad++
- Brackets
- Microsoft visual studio code

HTML VERSIONS

Since the early days of the web, there have been many versions of HTML:

HTML 1.0	provides basic version with a support for basic elements like text controls and images.
HTML 2.0	provides concept of forms, but they had basic tags like text boxes, button, etc.
HTML 3.2	supports for new form elements and for CSS (Cascading Style Sheet).
HTML 4.01	extends the support of CSS (external styling sheet emerged).
HTML 5	provides new form elements like input elements of different types.